

A New Interpretation of Information Rate

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ABSTRACT

Kelly asks how a bettor with private, imperfect advance knowledge of the outcome of a repeated chance event should size successive bets so that capital grows as fast as possible over many repetitions. Working from Shannon's definition of the transmission rate of a noisy channel, he shows that a bettor who stakes a fixed fraction of current capital on each round, chosen to maximize the expected value of the logarithm of wealth rather than the expected value of wealth itself, achieves an almost sure exponential growth rate that is bounded above and, under fair odds, exactly attained by the channel's rate of information transmission. The paper proceeds from a simple binary case to a general multi-symbol channel with arbitrary odds and, finally, to the case in which a fraction of every wager is lost to a track take. In each case the maximizing rule directs the bettor to apportion capital according to his true posterior probabilities of the underlying event, independent of the odds on offer, and never to bet the whole bankroll. The result gives a monetary, operational meaning to Shannon's abstract measure of information and is the origin of what later literature calls the Kelly criterion.

1 Introduction

John Larry Kelly, Jr. wrote this paper while working at Bell Telephone Laboratories, in the research group built around Claude Shannon. It is not, on its face, a finance paper. It is an attempt to answer a narrow question left open by Shannon's 1948 theory of communication: once a channel's rate of information transmission has been defined mathematically, in what sense, if any, does that quantity have value to someone who actually receives the transmitted symbols. **PAPER** Kelly's answer is that if the received symbols describe the outcome of a repeated wager on which fair or better than fair odds are available, a recipient who reinvests winnings and bets a constant fraction of current capital on each round can make personal wealth grow, almost surely, at an exponential rate exactly equal to the channel's information rate. The paper is short, entirely self-contained, and proceeds by direct calculation rather than by citing prior results in economics; it does not use the word utility except once, in the conclusion, to explicitly disclaim that framing.

The paper is included in this library because its central object, the fraction of capital that maximizes the long-run compound growth rate of wealth under uncertainty, is the direct ancestor of what the investment management literature calls the Kelly criterion, and because its method of proof (maximize the expectation of the logarithm of wealth, then invoke the strong law of large numbers to convert an expectation into an almost sure limit) is a template used throughout later growth-optimal portfolio theory.

2 Research Problem and Hypothesis

2.1 Research Question

Precisely stated: given a communication channel that repeatedly reveals, with some probability of error, the outcome of a chance event on which a bettor can wager at posted odds, what rule for choosing the fraction of capital staked on each round maximizes the exponential rate of growth of the bettor's capital as the number of rounds becomes large, and how does the resulting maximal growth rate relate to Shannon's rate of information transmission for that channel.

2.2 Motivation

Shannon's 1948 channel capacity theorem gives operational meaning to the transmission rate only through a coding theorem: with suitable encoding, binary digits can be sent at that rate with arbitrarily small error probability. That theorem is silent about any situation in which coding is not used or not possible, for example when the input to the channel is not a message to be decoded but the outcome of a real event, such as a horse race or, as Kelly notes explicitly in his final paragraph, the totality of inside information available to an

investor. **PAPER** Kelly wanted a second, independent justification for treating the transmission rate as economically meaningful, derived without appeal to coding at all. The question matters to a reader of this library because it is the first place the exponential growth rate of wealth, rather than its expectation, is proposed as the correct objective for repeated betting or investment, a proposal that recurs throughout portfolio theory whenever positions are held and reinvested over many periods.

2.3 Existing Literature at the Time

Two bodies of thought are in the background. First, Shannon (1948) had defined the rate of transmission of a noisy channel, R , in terms of entropies, and had shown that this quantity could be approached by suitable block coding with vanishing error probability, but had given it no significance in the absence of coding. Second, von Neumann and Morgenstern's expected utility theory (1944) offered the standard route by which economists attached a value measure to a random payoff, by way of a cost or utility function whose expectation is to be maximized. Kelly considers this second route explicitly and rejects it as a foundation for his problem: a cost function approach, he argues, is so general that it can be attached to almost any statistical situation and therefore sheds no particular light on what is distinctive about a communication system, namely that the receiver can profit from the received symbols. **PAPER**

2.4 Research Gap

What was missing, in Kelly's own framing, was an interpretation of the transmission rate that arose from the structure of a genuine communication-and-betting situation itself, without importing an arbitrary utility function from outside. No prior result connected the asymptotic growth rate of a reinvested capital stream to an information-theoretic quantity.

2.5 Contribution

The contribution is a constructive derivation, first for a binary symmetric channel and then for the general case, showing that the bettor who maximizes the expected logarithm of each round's capital growth attains, almost surely, an exponential growth rate equal to the mutual information between the transmitted and received symbols, which is exactly Shannon's rate R under fair odds, and strictly exceeds the no-channel growth rate by that same amount under any odds. The claimed contribution is fully demonstrated within the paper's own binary and finite-alphabet settings: the algebra is elementary and complete. It is narrower than later readers sometimes assume, however, since Kelly proves optimality only within the restricted class of strategies that bet a fixed, time-invariant fraction conditional on each received symbol; he states explicitly, in his closing paragraph, that whether such strategies dominate more general, history-dependent betting rules remains to be proved. **CRITIQUE**

3 Theoretical Framework and Mechanism

The paper is entirely theoretical, and the whole of its contribution lives in the mathematics, so this section reproduces the derivation in full rather than describing it in prose.

3.1 The Growth Rate Objective

Kelly defines the exponential rate of growth of the gambler's capital as

$$G = \lim_{N \rightarrow \infty} (1/N) \log_2(V_N / V_0) \quad (1)$$

V_N the gambler's capital after N bets.

V_0 starting capital.

$\log_2$ logarithm to base two, so that G is measured in bits per bet, the same unit as Shannon's transmission rate.

The choice of base two logarithm is not decorative: it is what makes G directly comparable, in the same units, to Shannon's channel capacity, and it is what will make the maximized G collapse onto the mutual information expression later in the derivation. Economically, G is the continuously reinvested, per-round compound growth rate of wealth: after N bets, $V_N = V_0 2^{NG}$. Maximizing G is equivalent to maximizing the geometric mean growth factor per bet, not the arithmetic mean capital after N bets, and the difference between these two objectives is the paper's central economic idea.

3.2 The Binary Noisy Channel

Consider a noisy binary channel in which each transmitted symbol is received correctly with probability q and incorrectly with probability p , where $p + q = 1$. The gambler bets a fixed fraction ℓ of current capital on the symbol he believes, from the received signal, to be correct, at even odds. After N bets, with W wins and L losses,

$$V_N = (1 + \ell)^W (1 - \ell)^L V_0 \quad (2)$$

By the strong law of large numbers, $W/N \rightarrow q$ and $L/N \rightarrow p$ almost surely, so that

$$G = q \log_2(1 + \ell) + p \log_2(1 - \ell) \quad (\text{with probability one}) \quad (3)$$

Equation (3) is the crux of the method: an almost sure, deterministic limit for G is obtained not by taking an expectation over V_N directly, which would be dominated by rare, extreme sample paths, but by taking a logarithm first (which converts the multiplicative process in equation (2) into a sum of independent, identically distributed terms) and only then applying the law of large numbers. This is exactly the technical move that makes log-wealth, rather than wealth itself, the tractable objective, and it is the same move used decades later in every growth-optimal or log-utility portfolio model.

MAXIMIZING G OVER THE BET FRACTION

Kelly maximizes G in equation (3) with respect to ℓ using the general fact that a weighted sum of logarithms $\sum X_i \log Y_i$, subject to $\sum Y_i = Y$, is maximized by setting Y_i proportional to X_i , a direct consequence of the concavity of the logarithm (Jensen's inequality). Applying this with $(1+\ell) \propto q$ and $(1-\ell) \propto p$, and using $q + p = 1$, gives

$$\ell^* = q - p \quad \text{and} \quad G_{\max} = 1 + p \log_2 p + q \log_2 q = R \quad (4)$$

Renaming q as the win probability, $\ell^* = q - p = 2q - 1$ is exactly the modern textbook Kelly fraction for an even-money bet with win probability q , and R is Shannon's expression for the capacity of a binary symmetric channel with crossover probability p . **INFERENCE** The equivalence $G_{\max} = R$ is the paper's headline result in its simplest case: the maximal achievable exponential growth rate exactly equals the information-theoretic capacity of the channel supplying the gambler's private knowledge, so information and wealth growth share a common unit, the bit.

3.3 The General Multi-Symbol Channel

Kelly then extends the argument to a channel with multiple, not necessarily equally likely, input symbols and independent noise, introducing the following notation.

$p(s)$ probability that the transmitted symbol is the s -th one.

$p(r/s)$ conditional probability that the received symbol is the r -th, given the transmitted symbol was the s -th.

$q(r)$, $q(s/)$ the received symbol probability and the posterior probability of the transmitted symbol r) given the received one.

α_s the odds paid on symbol s : dollars returned per dollar staked, including the stake.

$a(s/r)$ the fraction of capital the gambler stakes on symbol s after observing received symbol r .

With the gambler committing his full capital each round ($\sum_s a(s/r) = 1$, which entails no loss of generality since offsetting bets can hold money back), the growth rate is

$$G = \sum_{r,s} p(s,r) \log_2 [\alpha_s a(s/r)] \quad (5)$$

FAIR ODDS

Under fair odds, $\alpha_s = 1/p(s)$, which Kelly justifies as the odds toward which parimutuel or competitive betting markets are pushed by arbitrage, since a gambler could otherwise guarantee a profit by repeatedly betting an outcome priced above its true probability. Substituting $\alpha_s = 1/p(s)$ into equation (5) and splitting the sum,

$$G = \sum_{r,s} p(s,r) \log_2 [a(s/r) / p(s)] = \sum_{r,s} p(s,r) \log_2 a(s/r) + H(X) \quad (6)$$

where $H(X)$ is the entropy of the source, in Shannon's sense. The first term is maximized, again by the proportional-allocation argument, at $a(s/r) = p(s,r)/q(r) = q(s/r)$: the gambler should stake capital in exact proportion to his own posterior probability of each outcome given what he has observed. At that maximum,

$$G_{\max} = H(X) - H(X/Y) = R \quad (7)$$

which is Shannon's mutual information between transmitted and received symbols, his general definition of the rate of transmission. Equation (7) generalizes equation (4) from the binary case to an arbitrary finite alphabet: it is the paper's central theorem.

ODDS NOT FAIR, NO TRACK TAKE

If $\sum_s 1/\alpha_s = 1$ still holds (no operator's cut) but the individual α_s depart from $1/p(s)$, the same optimization gives $a(s/r) = q(s/r)$ again: the gambler ignores the posted odds entirely and bets according to his own true posterior probabilities, regardless of what the market is offering. **PAPER** The resulting maximal growth rate is

$$G_{\max} = H(\alpha) - H(X/Y), \quad \text{where } H(\alpha) = \sum_s p(s) \log_2 \alpha_s \quad (8)$$

Kelly shows $H(\alpha)$ is minimized, at value $H(X)$, exactly when odds are fair; any departure from fair odds in either direction (mispriced favorites or mispriced longshots) raises $H(\alpha)$ and therefore raises $G_{\max}$ above R , so a gambler with private information profits doubly, from the information itself and from any mispricing in the posted odds, and the two sources of profit are additively separable in equation (8).

THE TRACK TAKE

When a fraction of every dollar wagered is retained by the house, so $\sum_s 1/\alpha_s < 1$, the gambler can no longer costlessly hold money back via offsetting bets, and the problem becomes a constrained maximization with a genuine corner solution. Writing b_r for the fraction of capital held out of play after observing r ,

$$G = \sum_{r,s} p(s,r) \log_2 [b_r + \alpha_s a(s/r)] \quad (9)$$

$$\text{subject to } b_r + \sum_s a(s/r) = 1$$

Kelly solves this term by term for each r using Lagrange conditions, and shows the solution takes a threshold form: outcomes are ranked by $p(s)\alpha_s$, cumulative sums $p_t = \sum_1^t p(s)$ and $\sigma_t = \sum_1^t 1/\alpha_s$ are formed, and capital is staked only on the outcomes whose edge $p(s)\alpha_s$ exceeds a cutoff $F_t = (1-p_t)/(1-\sigma_t)$ taken at its minimum positive value; outcomes below the cutoff receive no stake at all, however positive their expected value in isolation. **PAPER** This is the paper's most economically consequential technical result and the one most directly continuous with later portfolio theory: with transaction costs, growth-optimal allocation is not smooth and continuous in the perceived edge; there is a positive minimum edge below which the optimal position is exactly zero, foreshadowing no-trade regions in transaction-cost portfolio optimization.

3.4 What the Framework Predicts

Stripped of the channel-coding language, the framework predicts three things that are directly testable against later theory and practice. First, a bettor or investor who reinvests and cares about long-run compound growth should size positions in proportion to true edge (the excess of true probability over break-even probability implied by the odds), not according to expected value maximization, which would prescribe betting everything on the single best-expected-value outcome. Second, more accurate private information translates linearly, through mutual information, into a higher attainable growth rate: better forecasts are worth a quantifiable amount of compounding. Third, transaction costs (the track take) do not merely shrink optimal position sizes; they eliminate participation altogether in marginal-edge opportunities. None of these predictions are tested empirically in the paper itself, a point taken up in Section 4.

4 Data and Empirical Strategy

The paper contains no data, no sample, and no empirical test of any kind. It is a closed-form mathematical derivation from stated axioms (a channel with known, stationary transition probabilities; a gambler who can bet any real fraction of capital; costless reinvestment). This is appropriate to its object and should not be read as a limitation of execution, but it is a real limitation of scope, and the review returns to it directly in Sections 7 and 8: every quantity in the model ($p(s)$, $p(r/s)$, α_s) is treated as known with certainty to the gambler, an assumption that has no empirical counterpart in real markets, where probabilities and edges must themselves be estimated from data and are never known exactly. **CRITIQUE**

5 Analytical Results and Illustrative Calculations

Because the paper offers no numerical results of its own, this section works two illustrative calculations, both using the paper's own binary-channel formulas from equations (3) and (4), to make the magnitudes and the paper's own qualifications concrete. These calculations are the reviewer's own, not the author's. **EXTENSION**

5.1 A Modest Edge, Compounded

Take $q = 0.6$, $p = 0.4$, a 60/40 informational edge on an even-money bet. From equation (4), the optimal fraction is $\ell^* = q - p = 0.2$: stake one fifth of capital each round. The maximal growth rate is

$$G_{\max} = 1 + 0.4 \log_2 0.4 + 0.6 \log_2 0.6 \approx 1 - 0.529 - 0.442 \approx 0.029 \text{ bits per bet.}$$

Over $N = 100$ rounds, capital grows by a factor of $2^{100 \times 0.029} \approx 2^{2.9} \approx 7.5$, roughly sevenfold, while never staking more than one fifth of capital on any single round and while losing 40 percent of individual rounds. The result illustrates the paper's core economic point: a small, repeatable statistical edge compounds into a large multiple only because the fraction staked is kept well below certainty-equivalent aggressiveness; the growth rate per round (0.029 bits) looks negligible in isolation and only becomes economically large through many repetitions, which is exactly the sense in which the paper's asymptotic, $N \rightarrow \infty$ framing matters and is not a rhetorical flourish.

5.2 Expected Value Maximization Versus Almost Sure Growth

Kelly is explicit that a gambler maximizing the expectation of V_N directly, rather than its logarithm, is driven to stake $\ell = 1$ (the whole bankroll) every round, since $\langle V_N \rangle = (2q)^N V_0$ is strictly increasing in ℓ up to $\ell = 1$. **PAPER** With $p = 0.4 > 0$, this strategy is, with probability one, eventually ruined: a single loss at $\ell = 1$ sets capital to zero, and a run of length N has probability $1 - (1-p)^N$ of containing at least one loss, which tends to 1 as N grows. The expected-value maximizer's own expectation is dominated entirely by the vanishingly small probability of a loss-free run, in which capital is astronomically large; almost every realized path ends at zero. This is not a subtlety of interpretation, it is the paper's explicit worked contrast, and it is the clearest demonstration in the paper of why maximizing $E[V_N]$ and maximizing the almost sure growth rate of V_N are different objectives with opposite prescriptions at the extreme.

5.3 A Reviewer's Extension: The Cost of a Wrong Belief

The paper's optimization is only ever performed with respect to the gambler's own probabilities; it says nothing about what happens when those probabilities are wrong, since it never entertains the possibility. **CRITIQUE** Suppose the true win probability is $q = 0.5$ (no edge exists) but the gambler believes $q = 0.6$ and, following equation (4), stakes $\ell = 0.2$ each round. The true expected growth rate, evaluated at the true probabilities but the wrongly optimized fraction, is

$$G = 0.5 \log_2(1.2) + 0.5 \log_2(0.8) \approx 0.5(0.263) + 0.5(-0.322) \approx -0.029 \text{ bits per bet,}$$

strictly negative: capital shrinks in expectation and, by the same law-of-large-numbers argument Kelly uses for the correct case, converges almost surely to zero. A rule that looks exactly like correctly applied Kelly sizing, given the gambler's stated beliefs, produces certain ruin once those beliefs are wrong by an amount that is not obviously large. **INFERENCE** This calculation is offered here as a bridge to Section 8 and is the single most important qualification later literature attaches to the paper's result: optimality is conditional on known probabilities, and the paper supplies no guidance, because it does not pose the question, on how sensitive G is to estimation error in $p(s)$ itself.

6 Extensions and Later Rigor

The paper contains no robustness section in the empirical sense; the closest analogue is to ask how well the 1956 derivation has held up against later, more rigorous treatment of the same problem, since this is a theoretical paper whose claims can be checked by proof rather than by resample.

6.1 Concern: Is Fixed-Fraction Betting Actually Optimal Among All Strategies?

Kelly proves optimality only within the class of strategies that bet a constant fraction $a(s/r)$ of current capital, conditional on the received symbol, on every round, and flags this restriction himself in his conclusion as an open question. **PAPER** Breiman (1961) closes this gap rigorously: for a sequence of independent, identically distributed favorable games, the log-optimal (Kelly) strategy asymptotically maximizes the growth rate of capital, and does so uniformly faster than any other essentially different strategy, in the sense that the ratio of Kelly capital to any other strategy's capital tends to infinity almost surely; Breiman further shows the Kelly strategy minimizes the expected time to reach a fixed, large capital target. Does this fully resolve the concern? Largely yes, for the i.i.d. setting Kelly himself analyzed; it does not extend the result to non-i.i.d., time-varying, or model-uncertain environments, which is precisely the setting of real financial markets. **CRITIQUE**

6.2 Concern: Does the Result Require the Betting Framing, or Is It More General?

Latané (1959), working independently in the finance literature and apparently unaware of Kelly's paper, proposed maximizing the expected geometric mean return, equivalently expected log wealth, as a criterion for choosing among risky investment ventures, on grounds of long-run capital accumulation rather than information theory. The convergence of two independent derivations, one from communication theory and one from investment choice under uncertainty, on the same log-wealth objective is itself evidence that Kelly's result is not an artifact of the communication-channel framing but a genuine property of repeated, reinvested multiplicative growth processes. **INFERENCE**

6.3 Concern: Does Log-Wealth Maximization Reduce to a Utility Function After All?

Kelly explicitly denies that his criterion rests on a value function attached to money; he motivates log-wealth maximization purely through the law of large numbers. Later authors, notably in the Merton-Samuelson continuous-time and static portfolio choice literature, show that maximizing expected log wealth each period is mathematically identical to maximizing expected utility under a constant relative risk aversion utility function with risk-aversion coefficient equal to one. **INFERENCE** Kelly's disclaimer is therefore true as a matter of motivation (he did not invoke Bernoulli's logarithmic utility, von Neumann-

Morgenstern axioms, or any declared risk preference) but not true as a matter of mathematical content: the derived optimal rule is observationally identical to log-utility maximization, and the paper's growth-rate criterion inherits every property, and every criticism, of that specific utility function, whether or not the author intended to adopt it. This distinction between motivation and mathematical content is not resolved anywhere in the paper and is best understood as an unacknowledged limitation rather than a genuine third alternative to utility theory. **CRITIQUE**

7 Critical Evaluation

ESTABLISHED

Within the class of fixed-fraction, conditional-on-signal betting strategies for a stationary, independent, known-probability channel, the strategy that maximizes expected log wealth per round attains, almost surely, the maximal exponential growth rate, and under fair odds this maximal rate exactly equals Shannon's mutual information rate. This is established by direct, checkable algebra within the paper and is not disputed anywhere in the subsequent literature.

SUPPORTED BUT CONDITIONAL

That fixed-fraction log-optimal betting is optimal among all conceivable betting strategies, not merely fixed-fraction ones, is supported, but only by Breiman's (1961) later proof under independent, identically distributed repetitions; Kelly's own paper leaves this open. The result is further conditional on the gambler's stated probabilities being the true probabilities; the paper supplies no result, and to the reviewer's knowledge the immediate follow-on literature of the 1960s supplied none either, for the case of estimated or uncertain probabilities.

SUGGESTIVE

The closing paragraph's suggestion that the channel of the theory might correspond to the totality of inside information available to an investor is suggestive of the paper's later application to financial markets, but the paper itself analyzes only a discrete, known-probability channel and does not attempt to characterize what a market's information rate would be, how it would be measured, or whether market odds behave as the fair or track-take odds the model assumes.

NOT ESTABLISHED

The paper does not establish, and does not attempt to establish, anything about finite-horizon performance, the variance or drawdown of the log-optimal strategy's capital path, robustness to mis-estimated probabilities, or behavior when successive bets are correlated rather than independent. All of these are exactly the properties that matter most to a practitioner applying the criterion with a finite time horizon and estimated, not known, parameters.

8 Strengths and Weaknesses

8.1 Strengths

- Self-contained, elementary derivation: every step follows from Jensen's inequality and the strong law of large numbers, so the results are fully verifiable by a reader with undergraduate probability, a strength Kelly himself would likely have recognized, since the paper is written to be checked step by step rather than taken on authority.
- Generality of the derivation: the same method carries, with no change of technique, from the simplest binary case through arbitrary finite alphabets, unfair odds, and a positive track take, which the reviewer identifies as a strength beyond what the paper itself claims for itself: the paper undersells how far a single Lagrangian argument travels across four successively harder betting environments.
- The track-take result (Section 3.3) anticipates, in a two-page argument, the qualitative shape of modern transaction-cost portfolio optimization: a no-trade region around the break-even edge. This connection is not drawn by Kelly, who had no reason to anticipate 1990s and 2000s market microstructure theory, and is identified here as an independent strength of the paper's reach.

8.2 Weaknesses

Known probabilities. Every probability in the model, $p(s)$, $p(r/s)$, and the fair-odds benchmark $1/p(s)$, is treated as exactly known to the gambler. Why it matters: real financial edges must be estimated from finite, noisy samples, and Section 5.3 shows that a modest error in the believed win probability can flip the true growth rate from positive to negative while the gambler, following the formula exactly, believes he is betting optimally. Potential bias: applying the Kelly formula with a plug-in probability estimate systematically overstates the safe bet size whenever the estimate itself is optimistic, because estimation error is not symmetric in its consequences for a strategy that is linear in the assumed edge. How it could be tested: simulate repeated application of the formula using probabilities estimated from rolling historical windows of finite length, and compare realized growth to the growth the formula would deliver under the population probabilities.

Independence across rounds. The law of large numbers argument underlying equations (3), (6), and (7) requires the sequence of outcomes to be independent (or at least to obey an ergodic theorem) across rounds. Why it matters: financial returns exhibit volatility clustering and serial dependence, so the almost sure convergence Kelly relies on may not hold, or may hold only at a much slower rate, in a market setting. Potential bias: applying fixed-fraction sizing under undiagnosed serial dependence can produce capital paths whose realized growth rate differs persistently, not just by finite-sample noise, from the theoretical G . How it could be tested: derive or simulate the growth-rate distribution under a Markov-dependent, rather than independent, outcome sequence and compare convergence speed and asymptotic bias to the i.i.d. benchmark.

Continuous divisibility and unlimited reinvestment. The model allows the gambler to stake any real-valued fraction of capital and to reinvest costlessly and instantly. Why it matters: real positions are subject to minimum lot sizes, margin and leverage constraints, and settlement frictions, so a computed ℓ^* that implies effective leverage above 1, which the general framework of Section 3.3 does not rule out for sufficiently favorable odds, may simply be infeasible to execute. Potential bias: the theoretical growth rate $G_{\max}$ is then an unattainable upper bound rather than a guide to feasible sizing. How it could be tested: resolve the constrained problem with an explicit upper bound on ℓ and compare the resulting growth rate to the unconstrained $G_{\max}$.

8.3 The Single Most Important Weakness

Of these, the known-probabilities assumption is the most consequential, because it is the one the paper's own applied legacy (bet and position sizing in real, uncertain markets) violates by construction: an investor never has Kelly's clean channel probabilities, only estimates of them, and Section 5.3's calculation shows the criterion is not gracefully robust to that gap, it can flip sign. **CRITIQUE**

9 Contradictions and Edge Cases

When does the mechanism fail outright? At $\ell = 1$ with any $p > 0$, Section 5.2 shows the mechanism does not merely underperform, it guarantees eventual ruin with probability one, even though it maximizes expected capital at every finite N . The paper's own criterion, maximize G rather than $\langle V_N \rangle$, exists precisely to avoid this failure mode, which makes the $\ell = 1$ case less an edge case than a built-in demonstration of why the paper's central methodological choice (logarithm before expectation) is necessary at all.

What happens at the boundary of the track-take result? Equation (9)'s threshold rule implies that as the track take grows, more and more nominally positive-edge outcomes fall below the cutoff F_t and receive zero stake. In the limit of a large enough take, every available bet can be priced below the cutoff and the model correctly prescribes not betting

at all, $b_r = 1$. This is a sensible, non-degenerate limit and, to the reviewer, a point in the model's favor: it does not mechanically prescribe positive stakes on positive-expected-value bets regardless of cost, unlike a naive expected-value rule.

Does the result survive outside a single, stationary channel? No channel-parameter drift is entertained anywhere in the paper. If $p(s)$ itself varies over the N rounds used to define the limit in equation (1), the almost sure convergence argument no longer applies unmodified, and G as defined is not obviously the right summary statistic; a time-varying edge would call for a time-varying ℓ_t , which the paper's fixed-fraction framing does not provide. **CRITIQUE** This is of first-order importance for any financial application, where edges (alpha, mispricing, informational advantage) are rarely constant, and it is the single largest distance between the 1956 channel model and a 2026 trading or investment context.

Could another mechanism generate the same $G_{\max} = R$ result? Yes: the equivalence in equation (7) is a restatement, in betting language, of the definition of mutual information; any setting in which a decision-maker's payoff is linear in the log of the posterior odds ratio will reproduce the same algebra. The result is therefore best understood as a general property of log-linear scoring under mutual information, of which betting is one instance, not a finance-specific or gambling-specific phenomenon. **INFERENCE**

10 Further Literature

Shannon (1948), A Mathematical Theory of Communication. Shared idea: the definition of the transmission rate R and the entropy quantities $H(X)$, $H(X/Y)$ that Kelly borrows wholesale. Relationship: Kelly's paper is explicitly a sequel, giving R an operational, betting-based interpretation that Shannon's own coding theorem does not supply. Why it matters: without Shannon's prior definitions the paper has no vocabulary; the review's equations (6) to (8) are unintelligible without it.

Von Neumann and Morgenstern (1944), Theory of Games and Economic Behavior. Shared idea: both are concerned with attaching a value measure to risky outcomes. Relationship: Kelly explicitly considers and rejects the expected-utility route as too general for his purposes, choosing instead a criterion derived from the structure of repeated, reinvested betting. Why it matters: the paper's identity as an alternative to, rather than an application of, expected utility theory is central to how later authors classify and critique it (Section 6.3).

Latané (1959), Criteria for Choice Among Risky Ventures. Shared idea: maximize expected geometric mean return / expected log wealth as the criterion for repeated investment choice. Relationship: an independent derivation, from finance rather than information theory, of the same objective function. Why it matters: cross-disciplinary convergence on log-wealth maximization is evidence the criterion reflects a real property of compounding, not an artifact of Kelly's communication-channel setup (Section 6.2).

Breiman (1961), Optimal Gambling Systems for Favorable Games. Shared idea: growth-optimal betting under repeated, independent favorable games. Relationship: supplies the general optimality proof, across all betting strategies and not merely fixed-fraction ones, that Kelly's own paper leaves as an open question. Why it matters: closes the paper's single largest stated gap (Section 6.1).

Samuelson (1979), Why We Should Not Make Mean Log of Wealth Big Though Years to Act Are Long. Shared idea: the log-wealth, Kelly-type growth criterion. Relationship: directly contradicts the practical implication, popular among some finance practitioners, that maximizing expected log wealth is the right rule for an investor with a long but finite horizon and any given degree of risk aversion; Samuelson shows the criterion is optimal only for an investor whose utility happens to be logarithmic, and that no amount of repetition makes it optimal for any other risk preference. Why it matters: it is the sharpest formal statement of the risk-preference critique implicit in Section 6.3 of this review, and it directly targets misapplications of Kelly's result rather than the result itself.

Merton (1969), Lifetime Portfolio Selection under Uncertainty: The Continuous-Time Case. Shared idea: optimal, continuously reinvested portfolio choice under a power or logarithmic utility function. Relationship: embeds Kelly's log-wealth special case inside a general constant relative risk aversion framework, showing exactly how much of the Kelly prescription (full commitment to the growth-optimal portfolio) is specific to risk-aversion coefficient one. Why it matters: gives the formal machinery for Section 6.3's claim that Kelly's rule is a particular utility function in disguise.

Thorp (2006) and MacLean, Thorp, and Ziemba (2011). Shared idea: direct application of the Kelly criterion to blackjack, sports betting, and equity markets, including fractional Kelly sizing as a practical response to parameter uncertainty. Relationship: the applied literature that carries Kelly's 1956 result into the setting this review's Section 8.2 identifies as the paper's main blind spot, estimation error, and proposes fractional Kelly (betting some fraction of ℓ^* , such as one half) as a partial, heuristic remedy. Why it matters: fractional Kelly is the practitioner's answer to exactly the wrong-belief calculation in Section 5.3, though it is a heuristic dampener rather than a formal solution to the estimation-error problem.

11 Research Gaps

11.1 Paper-Level Gaps

The paper leaves formally open whether fixed-fraction, conditional-on-signal strategies dominate all history-dependent betting strategies (later closed by Breiman for the i.i.d. case, Section 6.1); it offers no treatment of finite- N performance, only the $N \rightarrow \infty$ limit; and it offers no treatment of parameter uncertainty in $p(s)$ at all.

11.2 Literature-Level Gaps

Across the growth-optimal literature that follows Kelly, the reviewer identifies a persistent gap between the elegant asymptotic theory (Kelly, Breiman, and their continuous-time descendants) and practitioner heuristics such as fractional Kelly, which are used precisely because the formal theory does not, by itself, specify a principled sizing rule under estimated rather than known probabilities. A fully Bayesian treatment that derives, rather than assumes, a shrinkage factor analogous to fractional Kelly directly from posterior uncertainty over $p(s)$ would close this gap.

11.3 Methodological Gaps

Better identification of an investor's true edge, as opposed to a backward-looking sample estimate, would materially change how confidently the Kelly formula can be applied; better data would consist of out-of-sample, real-time forecast records rather than in-sample fitted probabilities; better estimation would draw on robust or shrinkage estimators explicitly designed to control the sign-flip failure mode demonstrated in Section 5.3.

12 Research Opportunities

1. Fractional Kelly Under Realistic Estimation Error

RESEARCH QUESTION: How much of the theoretical Kelly growth rate is preserved by half-Kelly and quarter-Kelly sizing once win probabilities must be estimated from a finite historical sample, rather than known exactly?

MOTIVATION: Section 5.3 shows full Kelly sizing is not robust to belief error; fractional Kelly is the standard practitioner response, but its benefit has rarely been quantified against the specific estimation-error magnitudes typical of real strategies.

HYPOTHESIS: The realized growth rate under fractional Kelly with estimated probabilities first rises and then falls in the fraction bet, with the growth-maximizing fraction strictly below one and decreasing in estimation noise.

ECONOMIC MECHANISM: Betting less than the plug-in optimum trades away some of the theoretical growth rate under correct beliefs for reduced sensitivity to the sign-flip failure mode of Section 5.3.

LITERATURE CONNECTION: Extends Kelly (1956) and Breiman (1961); formalizes the heuristic in Thorp (2006) and MacLean, Thorp, and Ziemba (2011).

DATA: Daily returns and rolling out-of-sample win-rate estimates for a systematic long-short equity or futures strategy with a documented signal, twenty or more years where available.

VARIABLES: Rolling estimated win probability, realized outcome, bet fraction, realized compound growth rate.

METHOD: Bootstrap resampling of the historical signal-outcome pairs under varying estimation-window lengths, applying equation (4) with the estimated rather than true probability.

IDENTIFICATION STRATEGY: Out-of-sample rolling estimation strictly separates the probability used for sizing from the outcome being sized against, avoiding look-ahead bias.

EXPECTED CONTRIBUTION: An empirically calibrated fractional-Kelly rule tied to the estimation-window length and signal turnover, rather than the conventional but ad hoc one-half fraction.

MAIN RISK: Regime change over a multi-decade sample could make the estimated relationship between fraction and growth unstable out of sample.

2. Multi-Asset Kelly Sizing With Transaction Costs and No-Short Constraints

RESEARCH QUESTION: How does Kelly's own track-take, threshold-based solution (Section 3.3, equation 9) generalize from a single sequential bet to a cross-section of simultaneously held assets with proportional transaction costs and short-sale constraints?

MOTIVATION: Kelly solves the single-asset, per-period track-take problem; real portfolios hold many assets at once, and the natural multivariate generalization has not, to the reviewer's knowledge, been derived directly from Kelly's own threshold logic rather than from the mean-variance tradition.

HYPOTHESIS: The multivariate solution retains a no-trade region generalizing equation (9), now defined over a set of assets rather than a single cutoff, and the region expands with the correlation among assets' edges.

ECONOMIC MECHANISM: Transaction costs create a fixed cost per position that must be compensated by expected log-growth contribution, exactly as in the single-asset case, but correlation among assets changes the marginal growth contribution of each additional position.

LITERATURE CONNECTION: Direct extension of Kelly (1956) Section 3.3; connects to Markowitz (1952) portfolio selection and to modern transaction-cost portfolio optimization.

DATA: Cross-section of liquid equity or futures returns with realistic bid-ask spread and commission estimates.

VARIABLES: Per-asset estimated edge, covariance matrix of returns, proportional transaction cost, position weight.

METHOD: Numerical constrained optimization of expected log portfolio growth subject to transaction costs and no-short constraints, benchmarked against the unconstrained log-optimal portfolio and against the mean-variance efficient frontier.

IDENTIFICATION STRATEGY: Not a causal identification problem; the exercise is a numerical and theoretical characterization, validated by simulation rather than by natural experiment.

EXPECTED CONTRIBUTION: A closed-form or numerically characterized no-trade region for multi-asset growth-optimal portfolios, directly traceable to Kelly's own single-asset derivation.

MAIN RISK: The optimization may lack a clean closed form outside special cases (for example, uncorrelated edges), limiting the result's generality.

3. Kelly Sizing Under Sequentially Correlated Bets in Prediction Markets

RESEARCH QUESTION: Does the asymptotic growth guarantee of equations (3) and (4) survive when successive bets are serially correlated, as in a sequence of related prediction-market contracts, rather than independent as Kelly assumes?

MOTIVATION: Section 9 identifies stationarity and independence as an untested boundary condition; prediction and sports-betting markets offer observable sequences of related wagers whose correlation structure can be measured directly.

HYPOTHESIS: Positive serial correlation among bet outcomes slows convergence of realized growth to the theoretical G and increases the probability of large drawdowns relative to the i.i.d. benchmark, for a fixed number of bets N .

ECONOMIC MECHANISM: Correlated outcomes reduce the effective number of independent trials underlying the law-of-large-numbers argument in equation (3), so the variance around the almost sure limit shrinks more slowly than $1/N$.

LITERATURE CONNECTION: Tests the independence assumption embedded in Kelly (1956) and Breiman (1961) against a documented real-world violation.

DATA: Historical odds and settlement outcomes from a betting exchange with a long, continuous order-book history for a single sport or event category.

VARIABLES: Implied probability from posted odds, realized outcome, estimated serial correlation of outcome residuals, realized versus theoretical growth rate.

METHOD: Estimate the serial dependence structure of outcome residuals, then simulate Kelly-sized betting sequences under the estimated dependence structure versus an i.i.d. null, comparing convergence speed and drawdown distributions.

IDENTIFICATION STRATEGY: Compare realized growth-rate paths against a bootstrap null that preserves marginal win probabilities but destroys serial dependence, isolating the effect of correlation from the effect of the edge itself.

EXPECTED CONTRIBUTION: An empirical measure of how much of the theoretical Kelly growth advantage survives realistic serial dependence in a genuine betting market.

MAIN RISK: Betting exchange data may have limited history or changing market structure over the sample, confounding the correlation estimate.

4. A Bayesian Kelly Criterion With Posterior-Derived Shrinkage

RESEARCH QUESTION: Can a formal, posterior-uncertainty-derived shrinkage factor replace the ad hoc one-half fractional Kelly heuristic used in practice?

MOTIVATION: Section 11.2 identifies the gap between Kelly's known-probability theory and practitioner heuristics for unknown probabilities; a Bayesian treatment can, in principle, derive rather than assume the correct degree of caution.

HYPOTHESIS: The Bayes-optimal bet fraction under a Beta posterior on the win probability is systematically below the plug-in Kelly fraction computed at the posterior mean, by an amount that shrinks as the number of observations underlying the posterior grows.

ECONOMIC MECHANISM: Posterior uncertainty about $p(s)$ adds a Jensen's-inequality penalty to expected log wealth that is absent from the point-estimate version of equation (4), because log is concave and averaging over posterior uncertainty in p lowers expected log growth relative to plugging in the posterior mean.

LITERATURE CONNECTION: Formalizes the estimation-error problem identified in this review's Section 8.2 and connects Kelly (1956) to Bayesian decision theory.

DATA: Simulated betting sequences with known data-generating probabilities, supplemented by one real dataset (for example, a systematic trading signal's historical hit rate) for external validity.

VARIABLES: Posterior mean and variance of the win probability, number of prior observations, Bayes-optimal bet fraction, realized growth rate.

METHOD: Derive the Bayes-optimal fraction by maximizing posterior expected log wealth under a Beta-Binomial model, then compare analytically and by simulation to plug-in Kelly and to conventional fractional Kelly.

IDENTIFICATION STRATEGY: Primarily a theoretical and simulation exercise; the real-data application uses out-of-sample validation rather than causal identification.

EXPECTED CONTRIBUTION: A principled, data-driven alternative to the conventional one-half fractional Kelly rule, with an explicit link to sample size.

MAIN RISK: The Beta-Binomial model may be too simple to capture real edge estimation, which often involves continuous, model-based probability forecasts rather than a stationary binomial process.

5. Do Professional Position Sizes Resemble Fractional Kelly?

RESEARCH QUESTION: Do position sizes chosen by professional discretionary or systematic traders scale with estimated edge in the manner Kelly's formula prescribes, and if not, what explains the deviation?

MOTIVATION: The paper's prescription (bet in proportion to true edge, ignore posted odds beyond their role in defining the edge) is a strong, falsifiable behavioral prediction that has rarely been tested against actual trade-level sizing decisions.

HYPOTHESIS: Realized position sizes are positively but sub-proportionally related to estimated edge, consistent with an implicit fractional-Kelly-like heuristic rather than either full Kelly or naive equal-weighting.

ECONOMIC MECHANISM: Career risk, drawdown constraints imposed by risk management, and aversion to the variance Kelly sizing permits would all push realized sizing below the theoretical optimum, consistent with Samuelson's (1979) critique that log-growth optimality does not match most investors' actual risk preferences.

LITERATURE CONNECTION: Tests Kelly (1956) and the fractional-Kelly practice documented in Thorp (2006) against revealed behavior, and connects to Samuelson's (1979) risk-preference critique.

DATA: Trade-level position size and contemporaneous model-implied edge from a proprietary trading desk or systematic fund willing to share anonymized records, or, failing that, publicly disclosed hedge fund position histories matched to a reconstructed signal.

VARIABLES: Estimated edge at trade initiation, realized position size as a fraction of capital, drawdown constraints in force, outcome.

METHOD: Regress realized position fraction on estimated edge and on risk-management constraints, testing the slope against the Kelly-implied slope of one.

IDENTIFICATION STRATEGY: Within-trader, within-strategy variation in estimated edge over time, holding the trader's risk-management regime fixed, isolates the sizing response to edge from cross-sectional differences in risk appetite.

EXPECTED CONTRIBUTION: Direct evidence on whether the Kelly criterion is a positive description of professional behavior or purely a normative benchmark practitioners deliberately under-apply.

MAIN RISK: Access to genuine trade-level sizing and edge-estimate data is the binding constraint; most desks treat this information as proprietary.

12.1 Ranking

Ranked by novelty, feasibility, data availability, methodological quality, economic importance, and potential contribution, the top three ideas are, in order, Idea 1 (fractional Kelly under estimation error), Idea 4 (Bayesian Kelly with posterior shrinkage), and Idea 2 (multi-asset Kelly with transaction costs). Idea 1 ranks first because it is the most directly feasible with widely available data, addresses the single weakness this review identifies as most consequential (Section 8.3), and would produce an immediately usable calibration for practitioners. Idea 4 ranks second because it offers the most methodologically novel contribution, replacing a heuristic with a derived rule, though it is more demanding to

execute convincingly and its real-data validation is harder to defend than Idea 1's. Idea 2 ranks third because it is the most direct mathematical extension of the paper's own most under-appreciated result (the track-take threshold rule), with clear economic importance for portfolio construction, but it is the least likely of the three to yield a clean closed-form contribution. Ideas 3 and 5 are valuable but rank lower here, principally on data availability: Idea 3 depends on betting-exchange data of uncertain length and quality, and Idea 5 depends on proprietary trade-level data that is difficult to obtain.

13 Learning Framework

13.1 Prerequisite Concepts

- Shannon entropy $H(X)$ and mutual information $H(X)$ minus $H(X \text{ given } Y)$, and their interpretation as the reducible uncertainty about X once Y is observed.
- The strong law of large numbers and the distinction between almost sure convergence and convergence in expectation.
- Concavity of the logarithm and the Jensen's-inequality argument used to solve each constrained maximization in the paper.
- Basic conditional probability notation, $p(s)$, $p(r/s)$, $q(s/r)$, as used throughout Section 3.

13.2 Core Concepts Learned

- The distinction between maximizing expected wealth and maximizing expected log wealth (equivalently, the almost sure growth rate of wealth), and why these two objectives can prescribe opposite actions at the extreme (Section 5.2).
- Growth-optimal (log-utility) betting or investing as a mathematically well-defined, testable criterion distinct from, though later shown equivalent to, a specific expected-utility function (Section 6.3).
- The idea that a positive expected edge is a necessary but not sufficient condition for taking a position once transaction costs are present (Section 3.3, equation 9).

13.3 Important Equations

Equations (1), (3), (4), (7), and (9) from Section 3 are the ones worth being able to reproduce from memory: the definition of the growth rate, the binary-channel growth rate and its maximizer, the general fair-odds result $G_{\max}$ equals R , and the track-take threshold rule.

13.4 Methods Learned

Maximizing a weighted sum of logarithms subject to a summation constraint via the proportional allocation identity (a special case of the Gibbs inequality / Jensen's inequality); converting a multiplicative, path-dependent process into an additive one by taking logarithms before applying a law of large numbers; solving a constrained per-period optimization by ranking outcomes and finding a cutoff, the same logical structure used in water-filling and other threshold-type resource allocation problems.

13.5 Questions to Test Understanding

COMPREHENSION

- In your own words, why does Kelly maximize the expectation of $\log V_N$ rather than the expectation of V_N itself, and what specifically goes wrong with the latter?
- State the fair-odds result $G_{\max}$ equals R in words, without symbols, and explain why the base of the logarithm in equation (1) matters for this equivalence to hold exactly.

ANALYSIS

- Reproduce the derivation of $l^* = q$ minus p from equation (3), explaining each use of the proportional-allocation identity.
- Explain why the track-take result (equation 9) produces a threshold rather than a smoothly shrinking position size as costs rise, and connect this to the concavity of the logarithm.

RESEARCH

- Design, at the level of a testable hypothesis, an experiment that would distinguish whether observed under-betting relative to full Kelly by real investors reflects parameter uncertainty (Idea 4) or risk aversion in Samuelson's sense (Idea 5), given that both predict smaller-than-Kelly positions.
- Using equation (8), explain under what conditions a gambler with no private information ($H(X \text{ given } Y)$ equal to $H(X)$) can still achieve a positive growth rate, and what this implies about separating the value of information from the value of favorable pricing in a market context.

13.6 Papers to Read Next

Breiman (1961); Latané (1959); Samuelson (1979); Merton (1969); Thorp (2006).

14 Conclusion

14.1 One-Sentence Thesis

A gambler or investor who reinvests over many rounds and cares about long-run compound growth should size positions to maximize expected log wealth, not expected wealth, and doing so under fair odds makes the achievable growth rate exactly equal to the information-theoretic value of the gambler's private knowledge.

14.2 Three Most Important Insights

- Maximizing expected wealth and maximizing the almost sure long-run growth rate of wealth are different objectives that can prescribe opposite actions, with expected-wealth maximization leading to certain ruin under repeated favorable but risky bets (Section 5.2).
- The maximal growth rate under fair odds is exactly Shannon's mutual information rate, giving a monetary, bit-denominated meaning to the value of private information (equation 7).
- Transaction costs create a genuine no-trade region, not merely smaller positions; a positive edge is necessary but not sufficient for participation once a track take exists (equation 9).

14.3 Strongest Evidence

Since this is a theoretical paper, its strongest evidence is the closed-form derivation itself: equations (3) through (9) are internally consistent, checkable line by line, and the special case $G_{\max}$ equals R in equation (4) can be verified by direct substitution, which is the closest analogue in a theoretical paper to a robust empirical result.

14.4 Biggest Weakness

Every probability the model requires is assumed known with certainty; Section 5.3's illustrative calculation shows the criterion is not gracefully robust to even modest errors in that assumption, and this is precisely the assumption real financial applications cannot satisfy.

14.5 Most Important Research Gap

A formal, posterior-uncertainty-derived sizing rule that replaces the ad hoc one-half fractional Kelly heuristic used throughout the applied literature (Section 11.2, Idea 4).

14.6 Best Research Opportunity

Idea 1, a data-driven calibration of fractional Kelly sizing against realistic, out-of-sample estimation error, ranked first in Section 12.1 for its combination of feasibility and direct relevance to the paper's central weakness.

14.7 Three Papers to Read Next

Breiman (1961), Optimal Gambling Systems for Favorable Games; Samuelson (1979), Why We Should Not Make Mean Log of Wealth Big Though Years to Act Are Long; Thorp (2006), The Kelly Criterion in Blackjack, Sports Betting, and the Stock Market.

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